IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - Data collection
  - Data wrangling
  - Exploratory Data Analysis with Data Visualization
  - Exploratory Data Analysis with SQL
  - Building an interactive map with Folium
  - Machine Learning Predictive analysis (Classification)
- Summary of all results
  - Exploratory Data Analysis results
  - Interactive analytics demo in screenshots
  - Predictive analysis results

Introduction

- Project background and context
  - SpaceX is the most successful company of the commercial space age, making space travel affordable. The company advertises Falcon 9 rocket launches on its website, with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. Based on public information and machine learning models, we are going to predict if SpaceX will reuse the first stage.
- Problems you want to find answers
  - How do variables such as payload mass, launch site, number of flights, and orbits affect the success of the first stage landing?
  - Does the rate of successful landings increase over the years? - What is the best algorithm that can be used for binary classification in this case?

Section 1

Methodology

Methodology

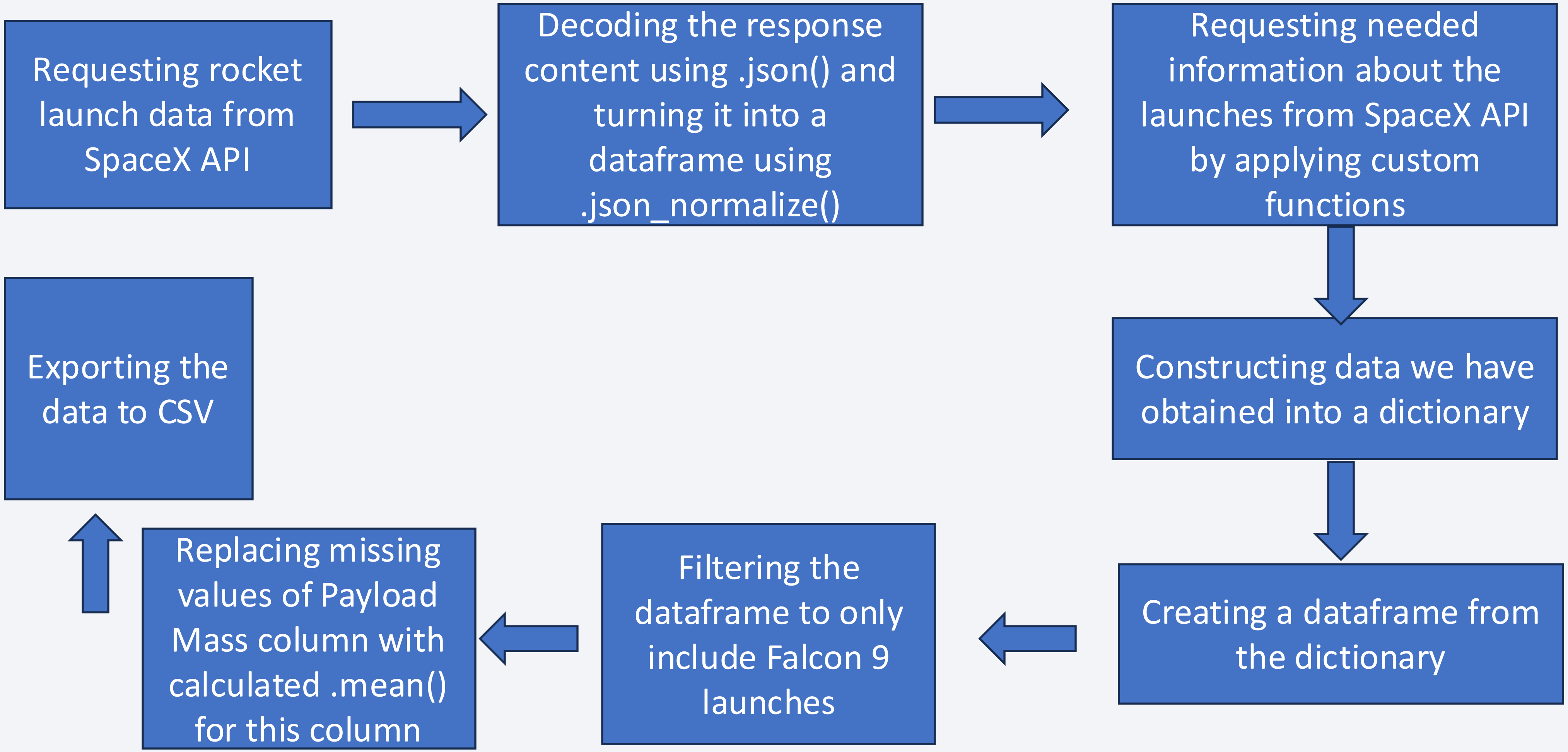
Executive Summary

- Data collection methodology:
  - Using SpaceX Rest API
  - Using Web Scrapping from Wikipedia
- Perform data wrangling
  - Filtering the data - Dealing with missing values - Using One Hot Encoding to prepare the data to a binary classification
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Data Collection

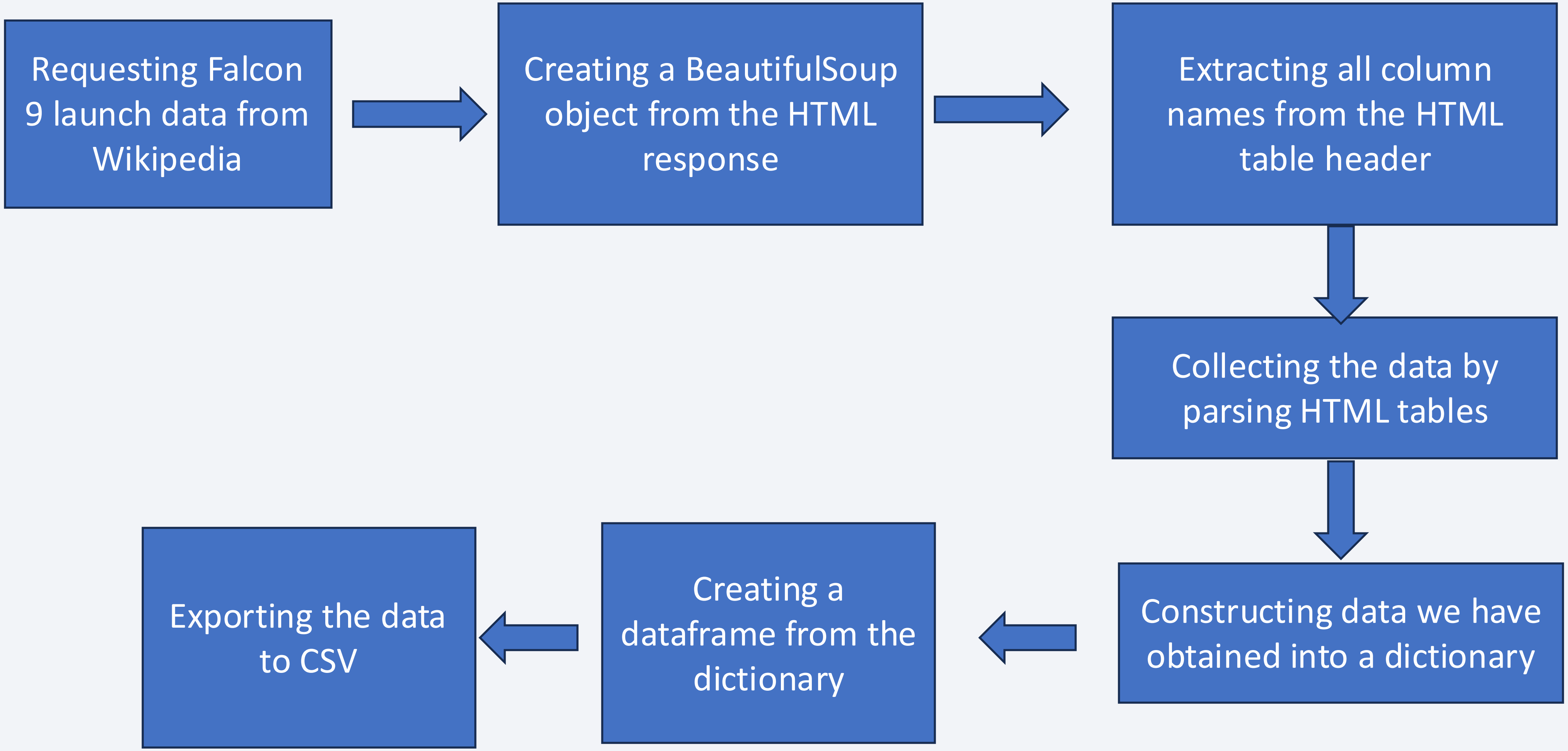
- Data collection process involved a combination of API requests from SpaceX REST API and Web Scraping data from a table in SpaceX’s Wikipedia entry. We had to use both of these data collection methods in order to get complete information about the launches for a more detailed analysis.
- Data Columns are obtained by using SpaceX REST API: FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude
- Data Columns are obtained by using Wikipedia Web Scraping: Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time

Data Collection – SpaceX API



Github URL: https://github.com/Lucky-Fan/Data_Science_Course/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Scraping



Github URL: https://github.com/Lucky-Fan/Data_Science_Course/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

In the dataset, booster landings are marked as successful (True) or unsuccessful (False) based on the landing type: **RTLS** (ground pad), **ASDS** (drone ship), or **Ocean** (sea landing). For example, True ASDS means a successful drone ship landing, while False RTLS indicates a failed ground pad landing.

Github URL: https://github.com/Lucky-Fan/Data_Science_Course/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

Perform exploratory Data Analysis and determine Training Labels

Calculate the number of launches on each site

Calculate the number and occurrence of each orbit

Calculate the number and occurrence of mission outcome per orbit typ

Create a landing outcome label from Outcome column

Exporting the data to CSV

EDA with Data Visualization

- Charts were plotted:

Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload Mass vs. Launch Site, Orbit Type vs. Success Rate, Flight Number vs. Orbit Type, Payload Mass vs Orbit Type and Success Rate Yearly Trend

- Scatter plots show the relationship between variables. If a relationship exists, they could be used in machine learning model.

- Bar charts show comparisons among discrete categories. The goal is to show the relationship between the specific categories being compared and a measured value. Line charts show trends in data over time (time series).

Github URL: https://github.com/Lucky-Fan/Data_Science_Course/blob/main/edadataviz.ipynb

EDA with SQL

Performed SQL queries:

- Displaying the names of the unique launch sites in the space mission
- Displaying 5 records where launch sites begin with the string ‘CCA’
- Displaying the total payload mass carried by boosters launched by NASA (CRS)
- Displaying average payload mass carried by booster version F9 v1.1
- Listing the date when the first successful landing outcome in ground pad was achieved
- Listing the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- Listing the total number of successful and failure mission outcomes • Listing the names of the booster versions which have carried the maximum payload mass
- Listing the failed landing outcomes in drone ship, their booster versions and launch site names
- Ranking the count of landing outcomes (such as Failure (drone ship) or Success (ground pad

Github URL: https://github.com/Lucky-Fan/Data_Science_Course/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- Markers of all Launch Sites:

- Added Marker with Circle, Popup Label and Text Label of NASA Johnson Space Center using its latitude and longitude coordinates.

- Added Markers with Circle, Popup Label and Text Label of all Launch Sites using their latitude and longitude coordinates to show their geographical locations and proximity to Equator and coasts.

- Coloured Markers of the launch outcomes for each Launch Site

- Distances between a Launch Site to its proximities

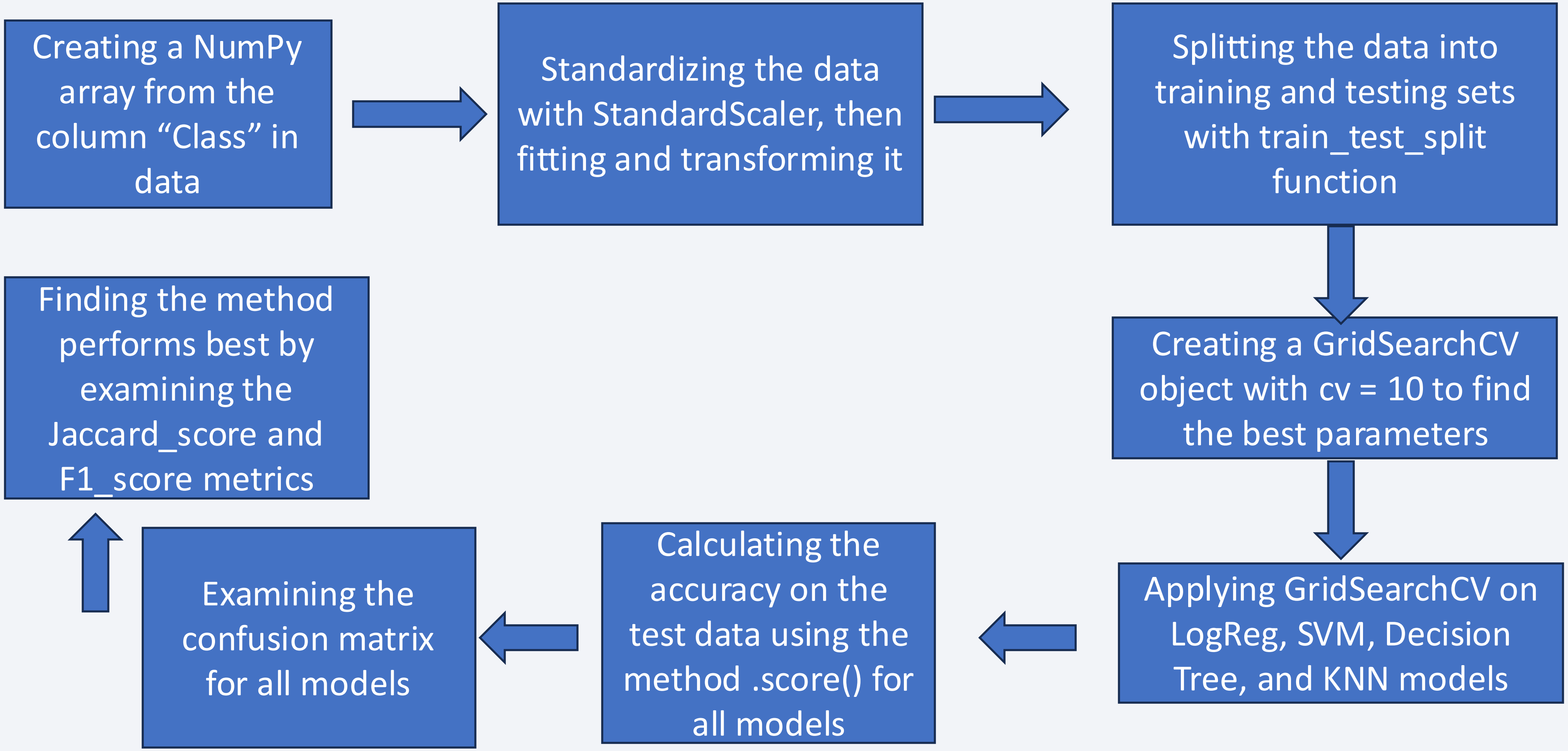
Github URL: https://github.com/Lucky-Fan/Data_Science_Course/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Launch Sites Dropdown List to enable Launch Site selection.
- Pie chart to show the total successful launches count for all sites and the Success vs. Failed counts for the site, if a specific Launch Site was selected.
- Added a slider to select Payload range.
- Scatter Chart of Payload Mass vs. Success Rate for the different Booster Versions

Github URL: https://github.com/Lucky-Fan/Data_Science_Course/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)



Github URL: https://github.com/Lucky-Fan/Data_Science_Course/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

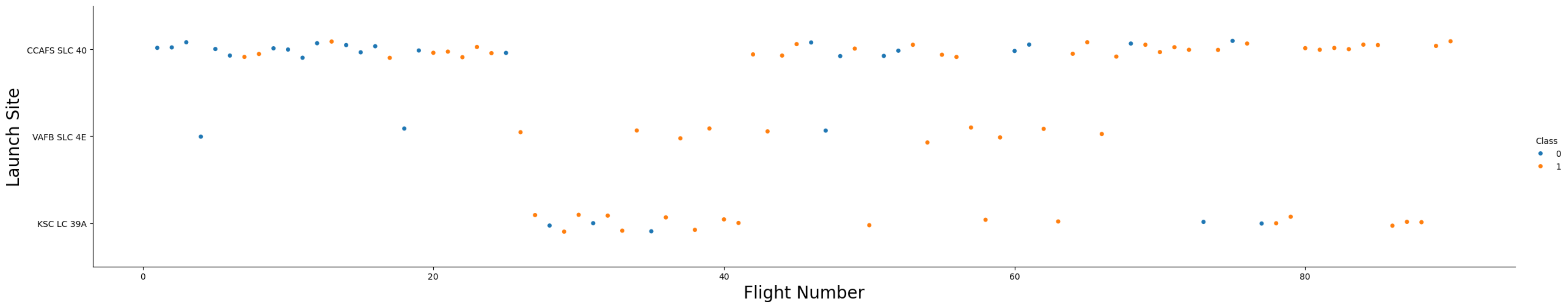
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

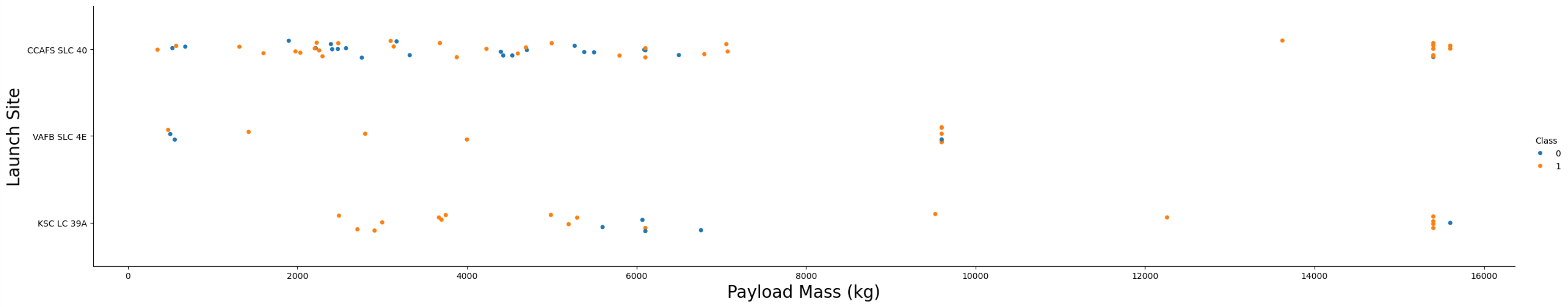
Flight Number vs. Launch Site



Explanation:

- About half of all launches occur at CCAFS SLC 40, while VAFB SLC 4E and KSC LC 39A show higher success rates.
- Overall, newer launches tend to have improved success rates.

Payload vs. Launch Site

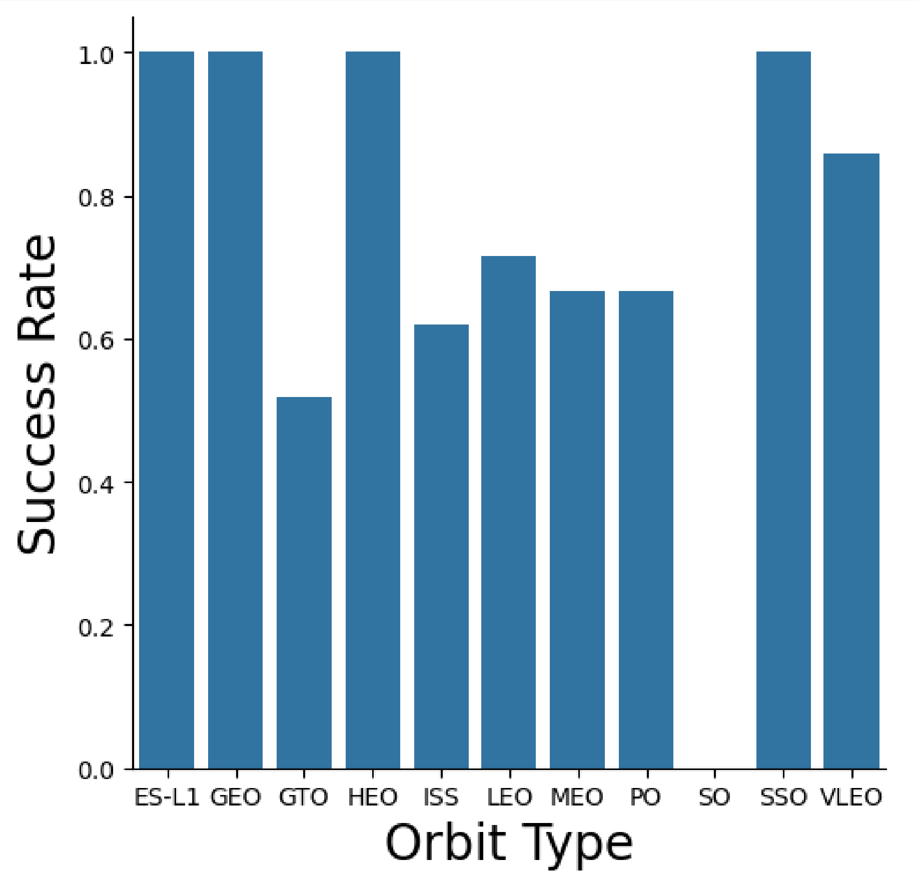


Explanation:

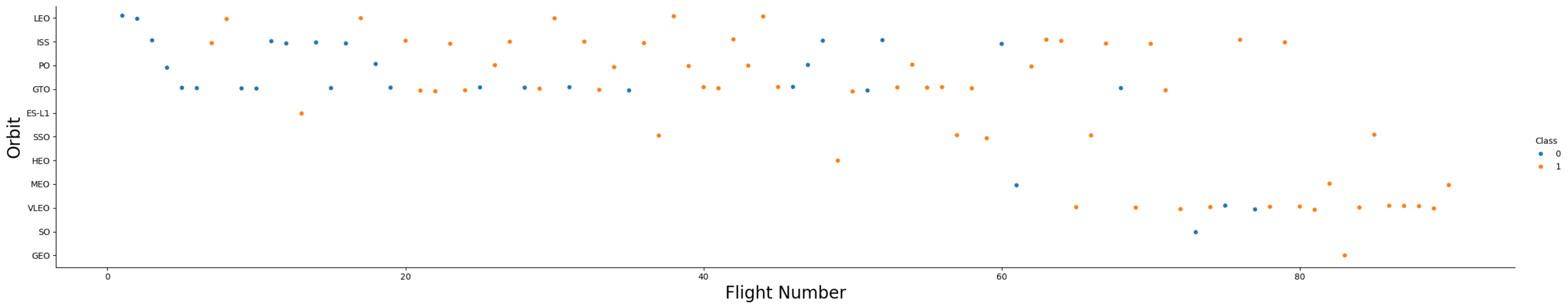
- Across all launch sites, higher payload mass is generally linked to higher success rates.
- Most launches exceeding 7000 kg were successful.
- Notably, KSC LC 39A achieved a 100% success rate for payloads under 5500 kg as well.

Success Rate vs. Orbit Type

- Explanation:
- Orbits with 100% success rate: ES-L1, GEO, HEO, SSO
- Orbits with 0% success rate: SO
- Orbits with 50–85% success rate: GTO, ISS, LEO, MEO, PO



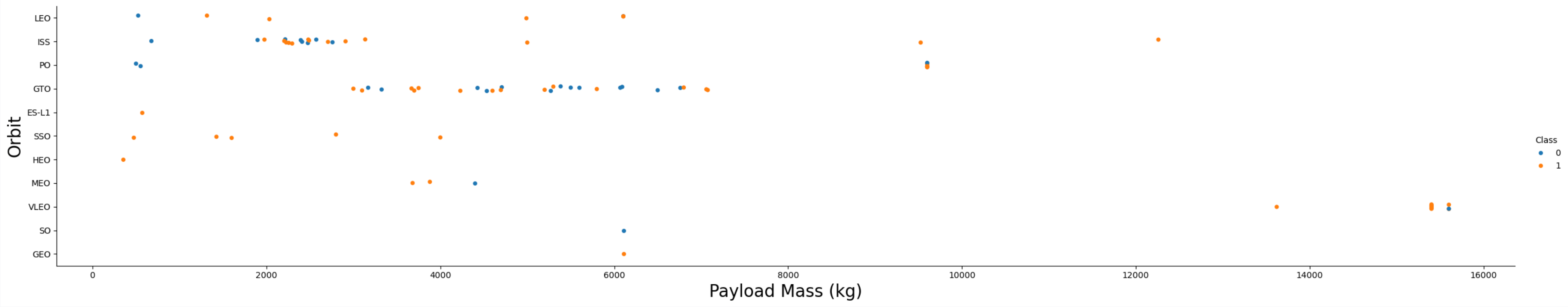
Flight Number vs. Orbit Type



Explanation:

- In LEO orbit, success rate tends to improve with more flights.
- However, in GTO orbit, flight number doesn't appear to impact success.

Payload vs. Orbit Type



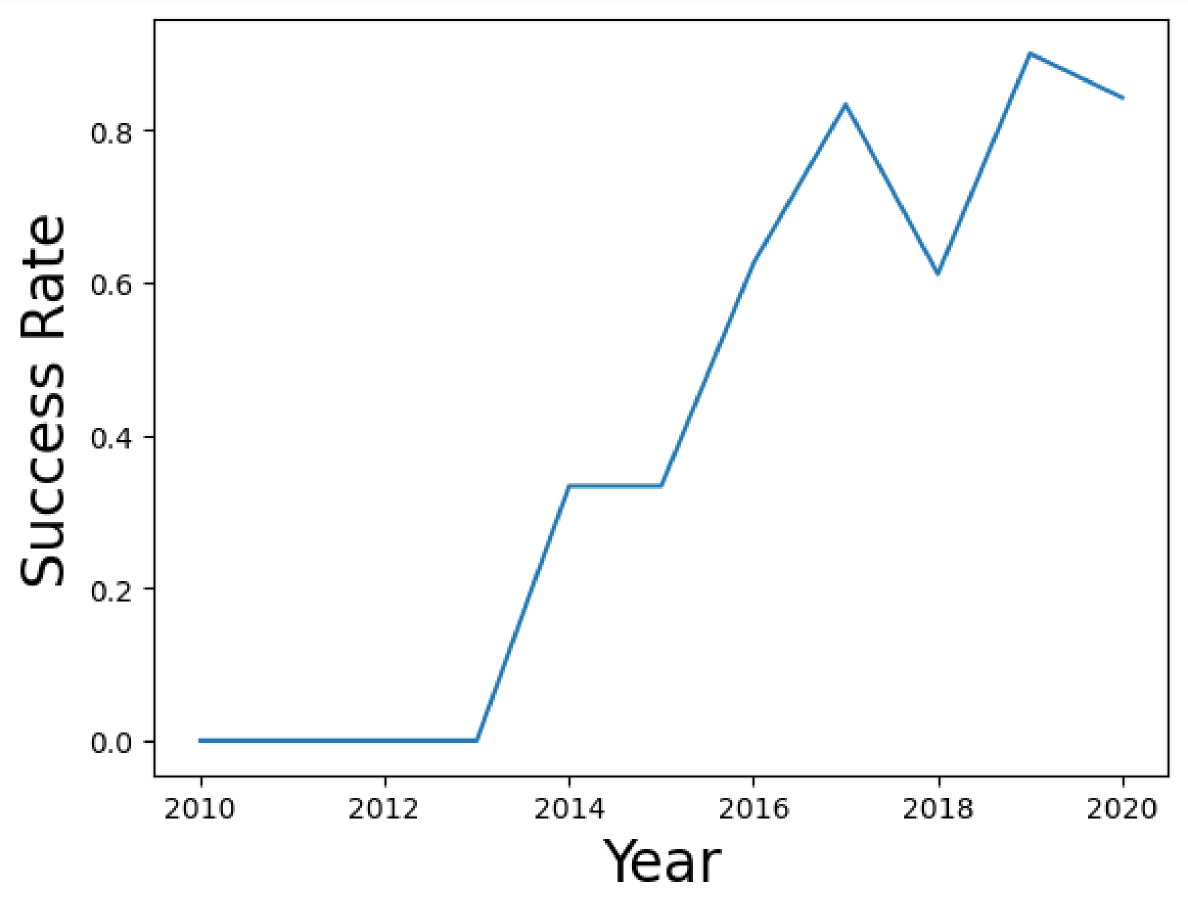
Explanation:

- Heavy payloads negatively impact GTO orbit success.
- But have a positive influence on Polar LEO (ISS) and LEO orbits.

Launch Success Yearly Trend

Explanation:

- The success rate since 2013 kept increasing till 2020.
- There is a dropdown in 2018 compared to 2017 and 2019.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
[12]: %sql select distinct launch_site from SPACEXTBL;

 * sqlite:///my_data1.db
Done.
```

[12]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation:

- Display the names of the unique launch sites in the space mission

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

[13]:
```
%sql select * from SPACEXTBL where launch_site like 'CCA%' limit 5;
```

```
 * sqlite:///my_data1.db
Done.
```

[13]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Explanation:

- Displaying 5 records where launch sites begin with the string 'CCA'.

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

[14]:
```
%sql select sum(payload_mass__kg_) as total_payload_mass from SPACEXTBL where customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

[14]:

total_payload_mass
45596

Explanation:

- Displaying the total payload mass carried by boosters launched by NASA (CRS).

Average Payload Mass by F9 v1.1

Task 4 ¶

Display average payload mass carried by booster version F9 v1.1

```
[15]: %sql select avg(payload_mass__kg_) as average_payload_mass from SPACEXTBL where booster_version like '%F9 v1.1%';
```

```
 * sqlite:///my_data1.db
Done.
```

[15]:

average_payload_mass
2534.6666666666665

Explanation:

- Displaying average payload mass carried by booster version F9 v1.1.

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
[24]: %sql select min(date) as first_successful_landing from SPACEXTBL where Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

[24]:

first_successful_landing
2015-12-22

Explanation

- List the date when the first successful landing outcome in ground pad

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

[25]:
```
%sql select booster_version from SPACEXTBL where Landing_Outcome = 'Success (drone ship)' and payload_mass__kg_ between 4
```

```
 * sqlite:///my_data1.db
Done.
```

[25]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Explanation:

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

[26]:
```
%sql select mission_outcome, count(*) as total_number from SPACEXTBL group by mission_outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

[26]:

Mission_Outcome	total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Explanation:

- Calculate the total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
[27]: %sql select booster_version from SPACEXTBL where payload_mass__kg_ = (select max(payload_mass__kg_) from SPACEXTBL);
```

```
 * sqlite:///my_data1.db
Done.
```

[27]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Explanation:

- List the names of the booster which have carried the maximum payload mass

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

[33]:
```
%%sql select strftime('%m', date) AS month, date, booster_version, launch_site, Landing_Outcome from SPACEXTBL
      where Landing_Outcome = 'Failure (drone ship)' and strftime('%Y', date) = '2015';
```

```
 * sqlite:///my_data1.db
Done.
```

[33]:

month	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Explanation:

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

[34]:
```
%%sql select Landing_Outcome, count(*) as count_outcomes from SPACEXTBL
      where date between '2010-06-04' and '2017-03-20'
      group by Landing_Outcome
      order by count_outcomes desc;
```

```
 * sqlite:///my_data1.db
Done.
```

[34]:

Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

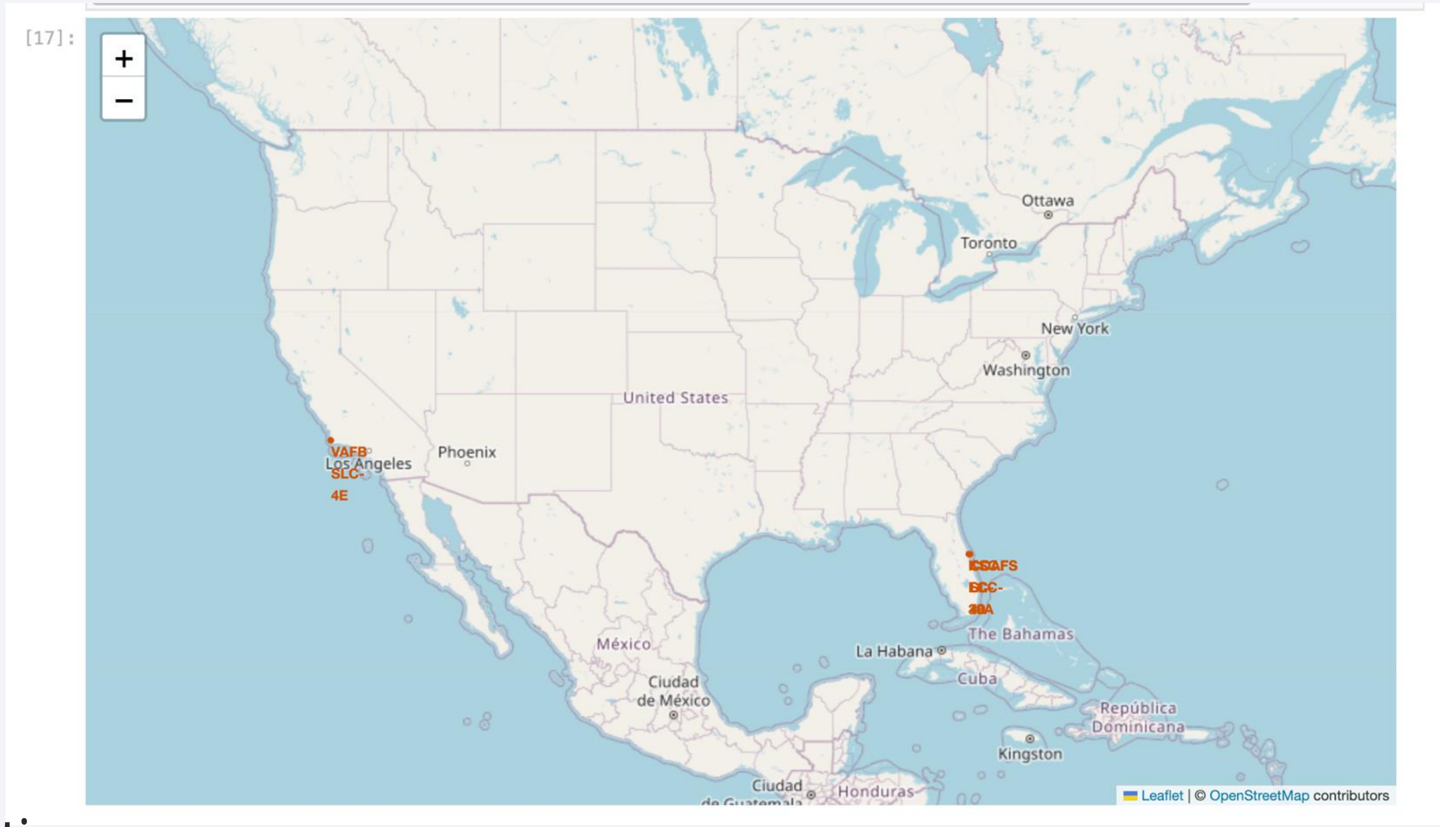
Explanation:

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

Section 3

Launch Sites Proximities Analysis

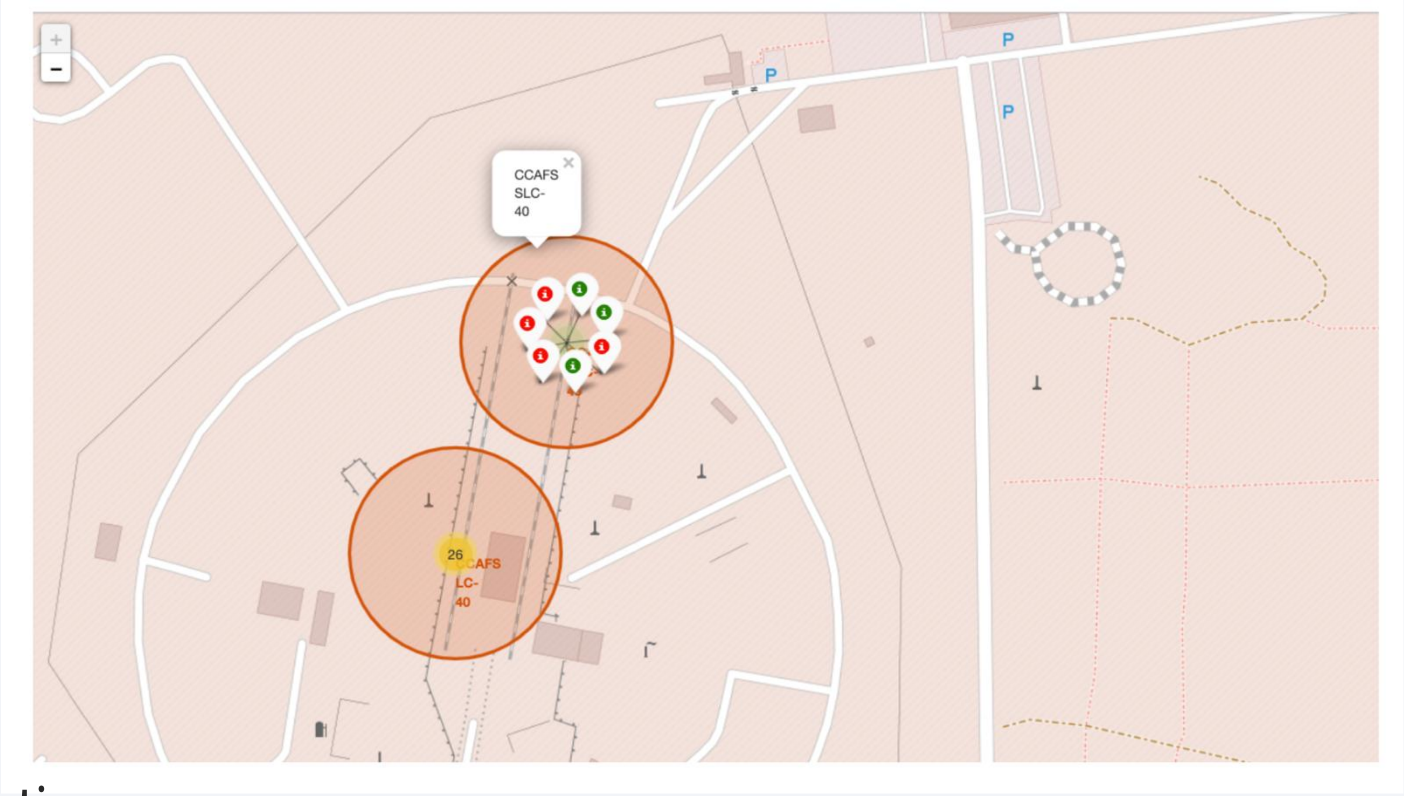
All launch sites' location markers on a global map



Explanation:

- Most launch sites are near the equator, where Earth's rotation (1670 km/h) gives rockets an extra speed boost due to inertia. They're also located near coasts to reduce the risk of debris falling near populated areas.

Color-labeled launch outcomes on the map



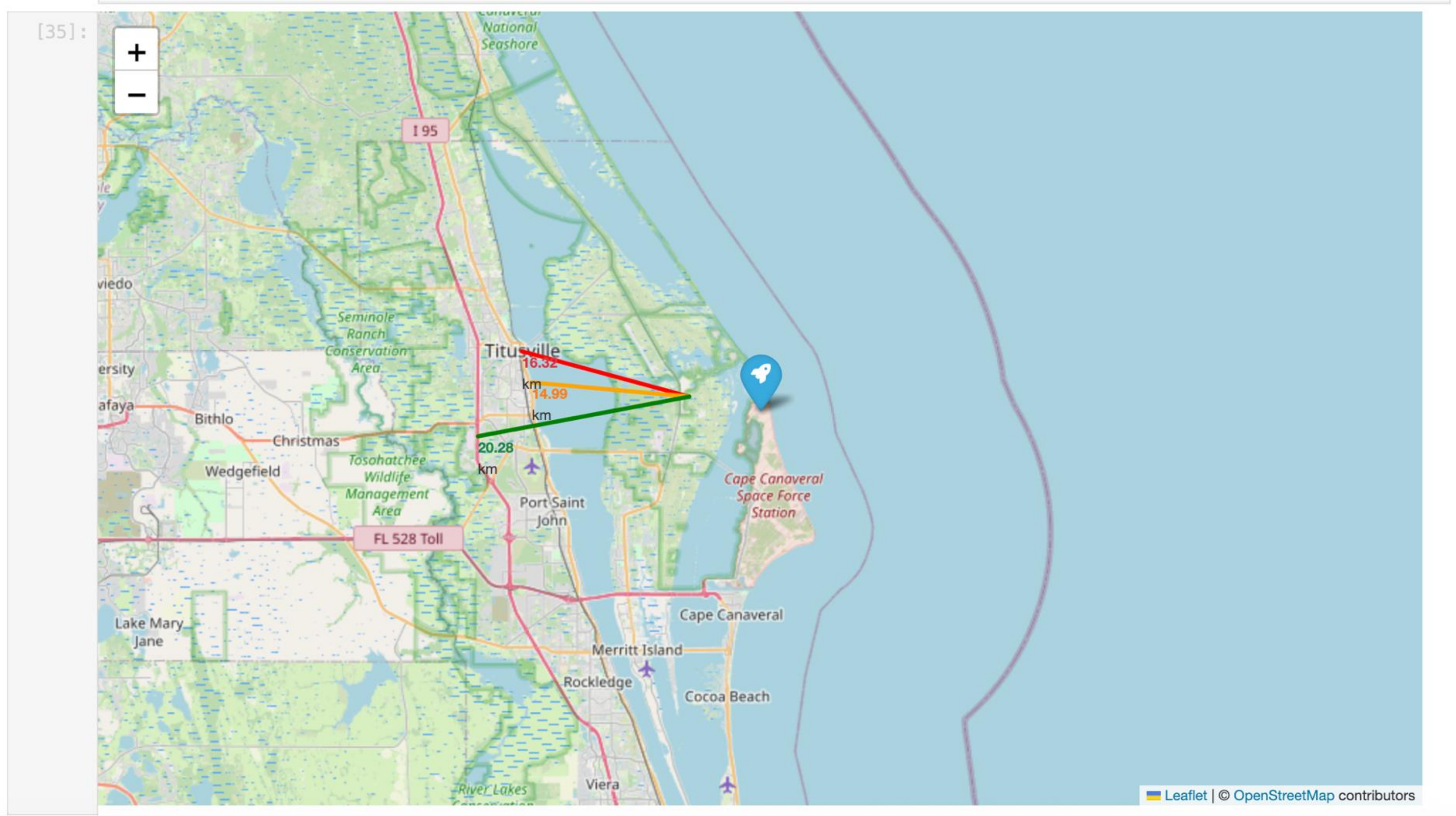
Explanation:

- Colour-labeled markers identify which launch sites have relatively high success rates (Green-Successful, Red-Failed).
- Launch Site CCAFS SLC-40 has a high failure Rate.

Selected launch site to its proximities

Explanation:

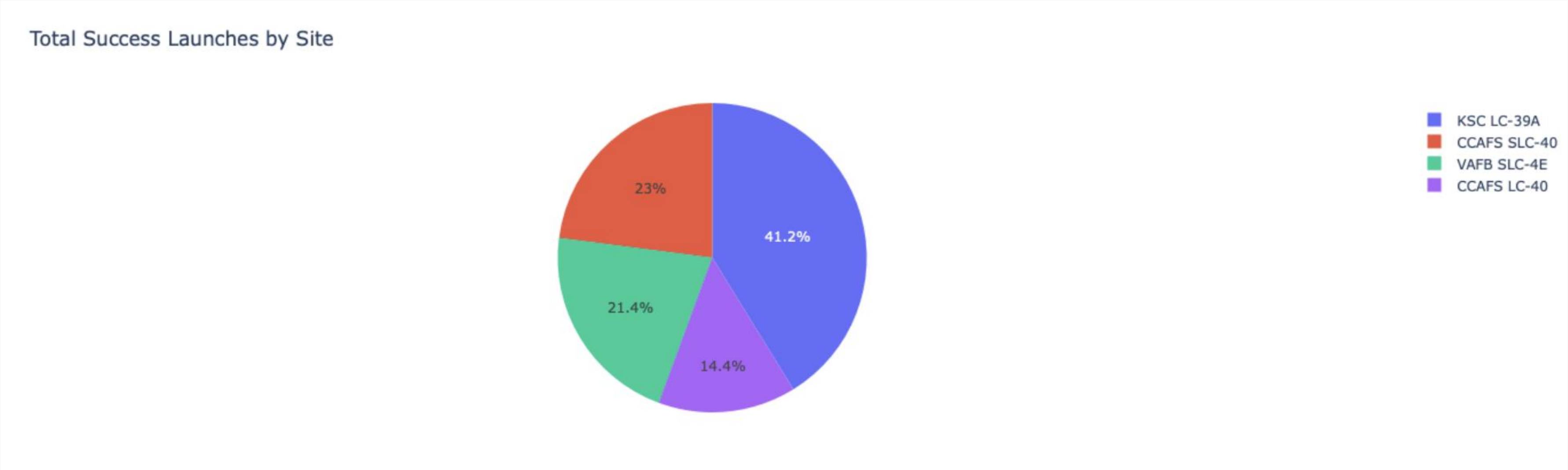
- Relative close to railway (16.32 km)
- Relative close to highway (20.28 km)
- Relative close to coastline (14.99 km)



Section 4

Build a Dashboard with Plotly Dash

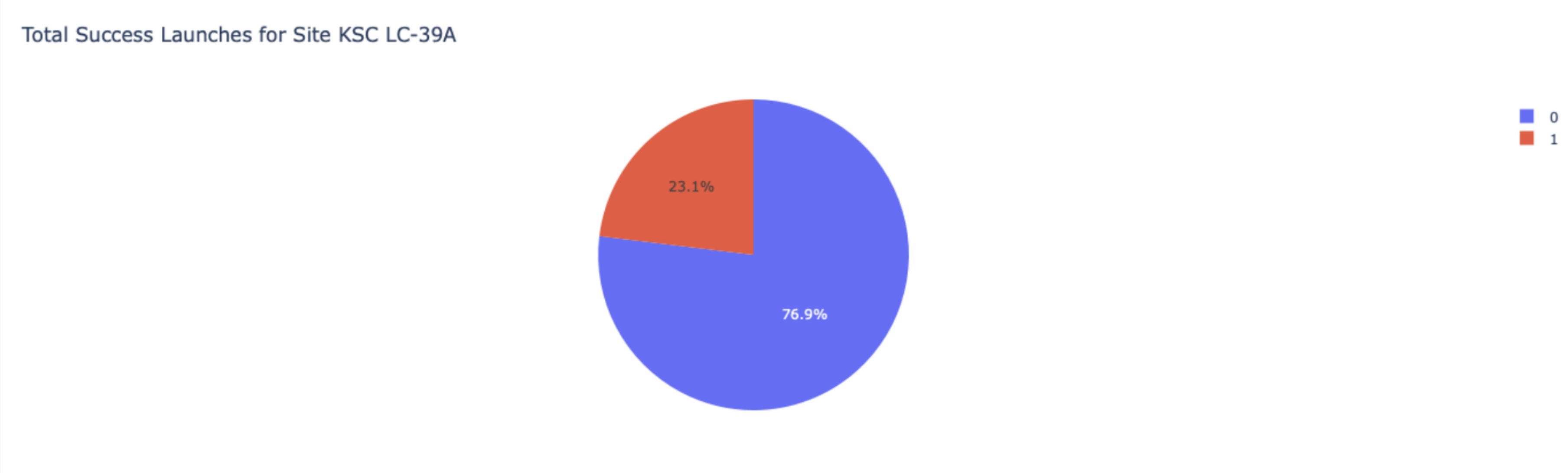
Launch success count for all sites



Explanation:

- From all the sites, KSC LC-39A has the most successful launches.

Launch site with highest launch success ratio



Explanation:

- KSC LC-39A has the highest launch success rate (76.9%).

Payload vs. Launch Outcome scatter plot for all sites

Explanation:

- Payloads between 2000 and 5500 kg have the highest success rate.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

Test dataset

[40]:

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.800000	0.800000	0.666667	0.800000
F1_Score	0.888889	0.888889	0.800000	0.888889
Accuracy	0.833333	0.833333	0.722222	0.833333

Whole dataset

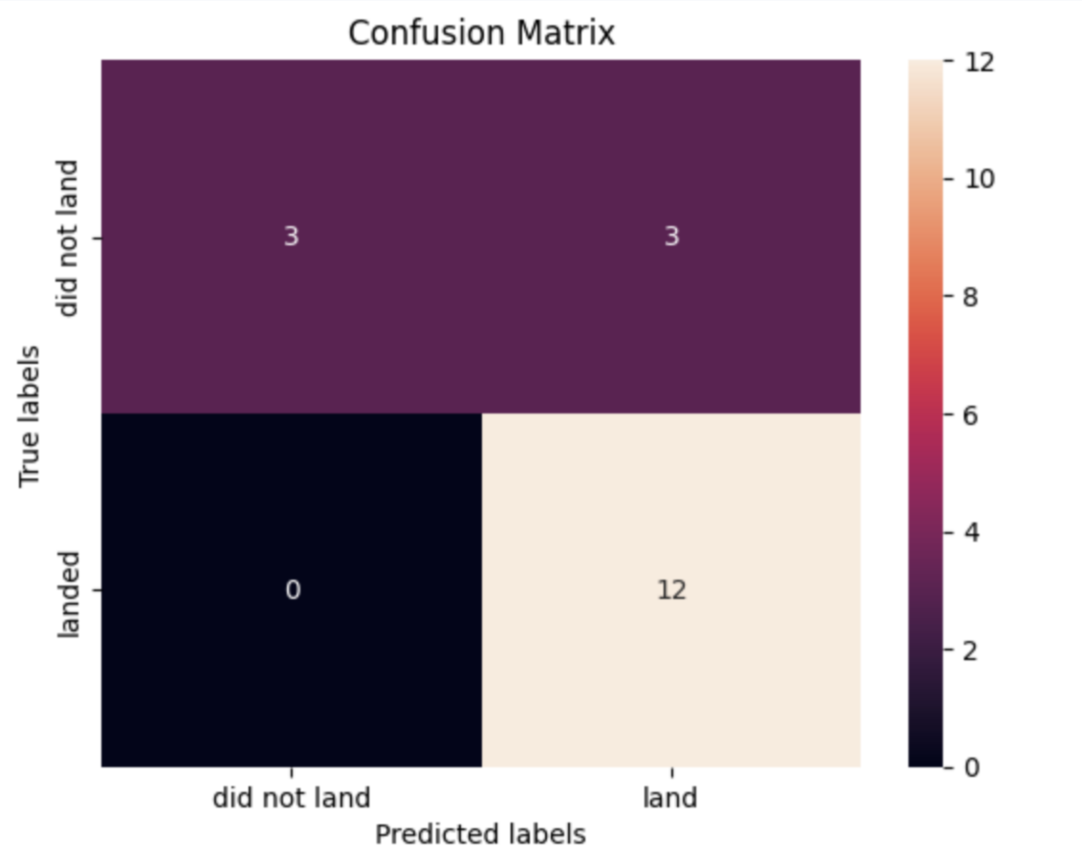
[41]:

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.833333	0.845070	0.850746	0.819444
F1_Score	0.909091	0.916031	0.919355	0.900763
Accuracy	0.866667	0.877778	0.888889	0.855556

Explanation:

- From the whole dataset result, we can find the best model is Decision tree model.

Confusion Matrix



Examining the confusion matrix, we see that logistic regression can distinguish between the different classes. We see that the problem is false positives.

Conclusions

- The **Decision Tree model** performs best on this dataset.
- **Lower payload mass** is associated with higher launch success.
- Most launch sites are near the **equator** and **coastlines**, optimizing speed and safety.
- **Launch success rates** have improved over time.

Thank you!